Day 14 Masterclass

How to Generate Research Ideas From Everyday Problems

প্রতিদিনের সমস্যা থেকে গবেষণার আইডিয়া তৈরি করার সহজ পদ্ধতি

For all disciplines | Beginner friendly | Live practice



Salek Data Lab

What You Will Learn

এই ক্লাস শেষে আপনি কী করতে পারবেন

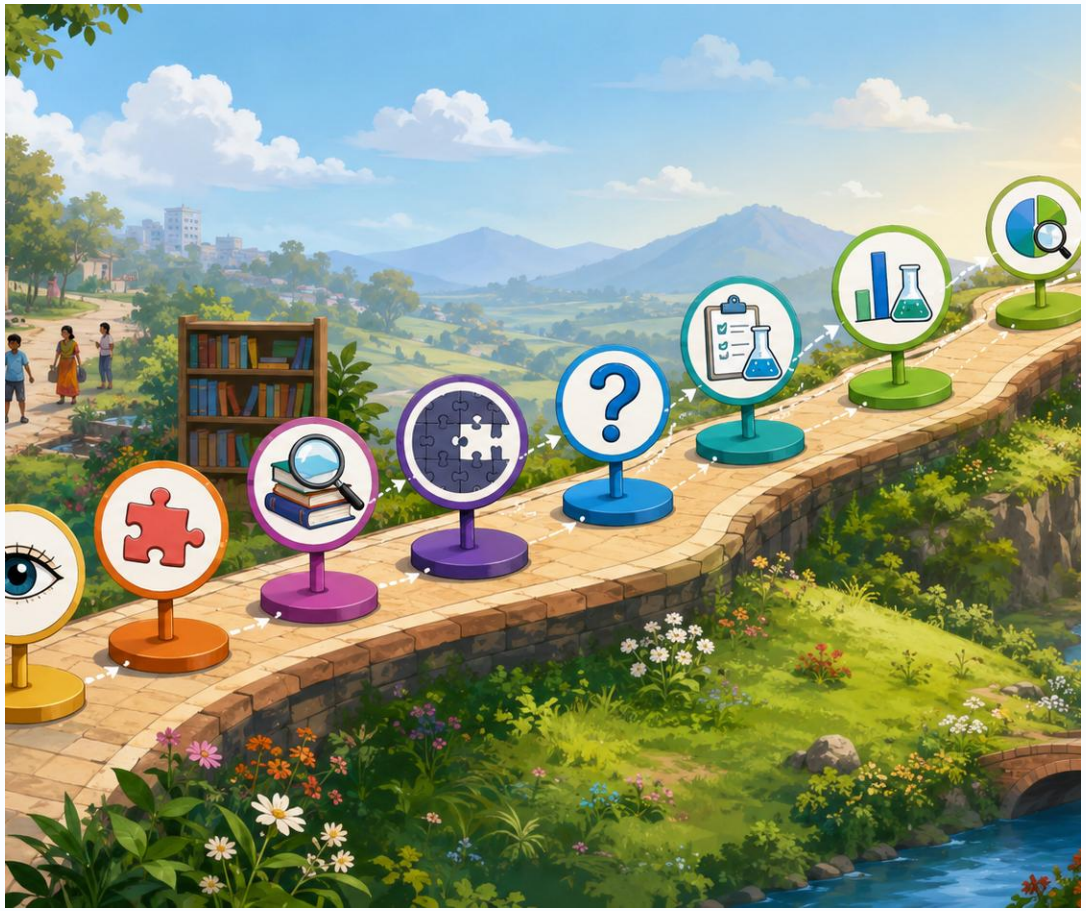
1. **Observe problems**
   সমস্যা লক্ষ্য করা
2. **Ask researchable questions**
   গবেষণাযোগ্য প্রশ্ন করা
3. **Check evidence**
   আগের গবেষণা দেখা
4. **Select feasible ideas**
   বাস্তবসম্মত আইডিয়া বাছাই
5. **Write clear topic titles**
   পরিষ্কার research title লেখা

Main promise: You do not need to wait for a perfect idea. You need a system to turn everyday problems into researchable questions.

মূল কথা: perfect idea আসার অপেক্ষা নয়; একটি practical system ব্যবহার করে idea বানাতে হবে।

Research Ideas Are Hidden in Problems

সমস্যার ভেতরেই research idea থাকে



Everyday problem

বাস, হাসপাতাল, classroom, farm, market, app, language use - সব জায়গায় problem আছে।

Research lens

একজন researcher problem দেখে প্রশ্ন করে: কেন, কীভাবে, কার ওপর, কতটা, কোন প্রমাণে?

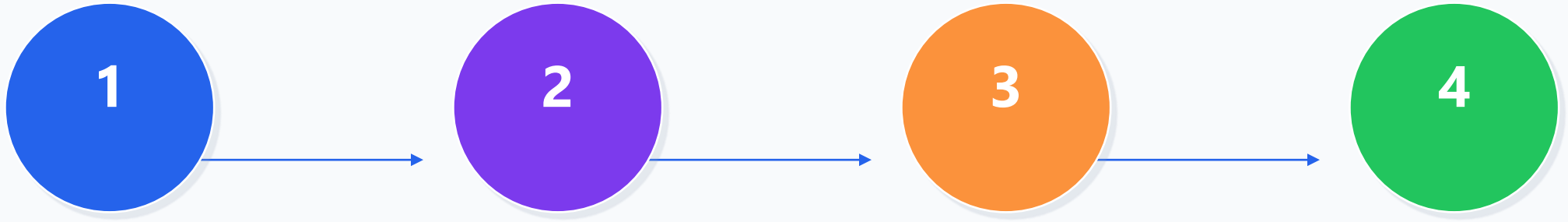
Research idea

Problem যখন specific question, evidence এবং method পায়, তখন সেটি research idea হয়।

Formula: Problem + Curiosity + Evidence + Method = Research Idea

The OAIT Framework

Observe, Ask, Investigate, Transform



Observe

দেখুন

What is happening?

Ask

প্রশ্ন করুন

Why or how?

Investigate

প্রমাণ খুঁজুন

What is already known?

Transform

রূপ দিন

Make a clear topic

Use this framework for any discipline: English, CSE, MBBS, Business, Agriculture, Engineering, Law and Social Science.

Where Do Ideas Come From?

আইডিয়ার ৮টি উৎস

Classroom

শিক্ষক-শিক্ষার্থী সমস্যা

Hospital

patient care ও adherence

Technology

apps, AI, cyber issues

Workplace

productivity ও management

Language

writing, translation, discourse

Community

social problems ও policy

Environment

climate, water, farming

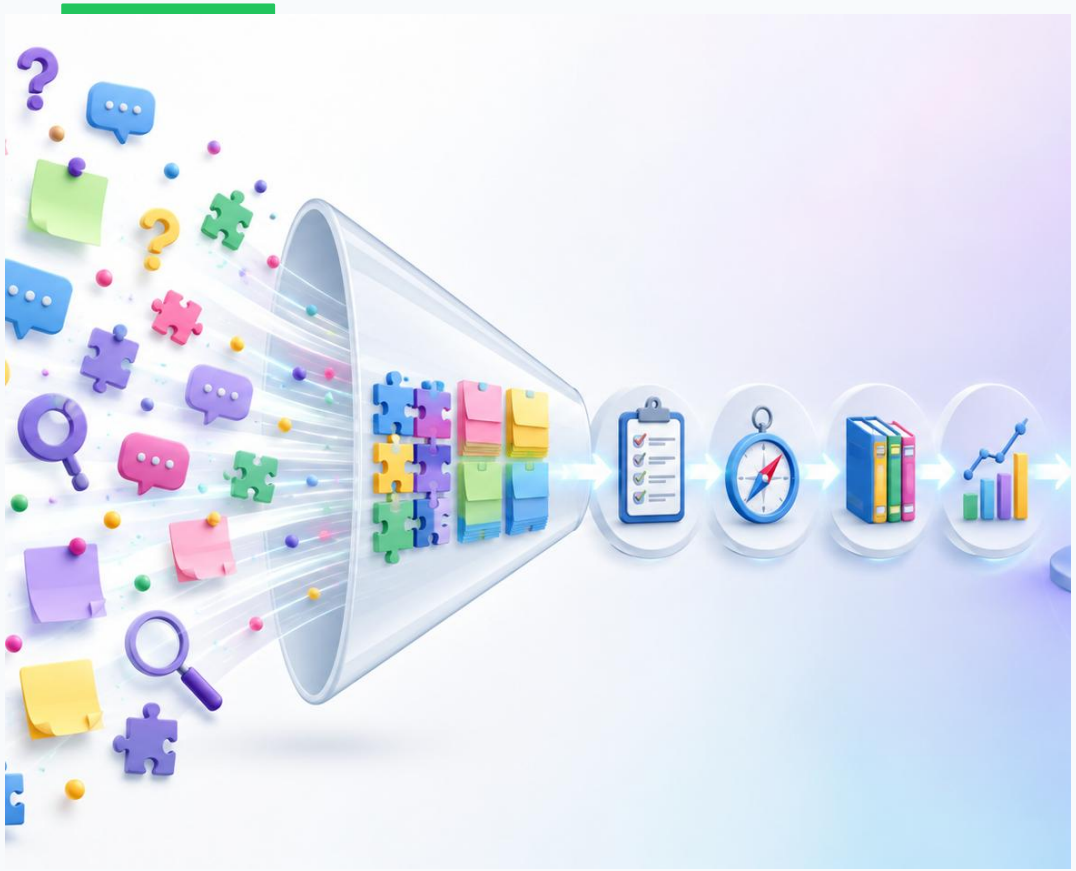
Literature

texts, culture, identity

Practice: keep an Idea Diary for 7 days. Write at least 5 problems from your own discipline.

From Problem to Research Problem

সাধারণ সমস্যা থেকে গবেষণার সমস্যা



Everyday problem

Students use AI tools.

Research problem

We do not know how AI feedback changes revision behaviour.

Evidence need

Compare drafts, interviews, or survey responses.

Possible method

Mixed method or document analysis.

Rule: a research problem must show uncertainty. Something is unknown, unclear, underexplored or debated.

Ask Better Questions With 5W1H

ভালো প্রশ্ন করার সহজ টুল

Example

Problem: students struggle with academic writing.

Better questions: Who struggles most? Why? What support helps? Which discipline?

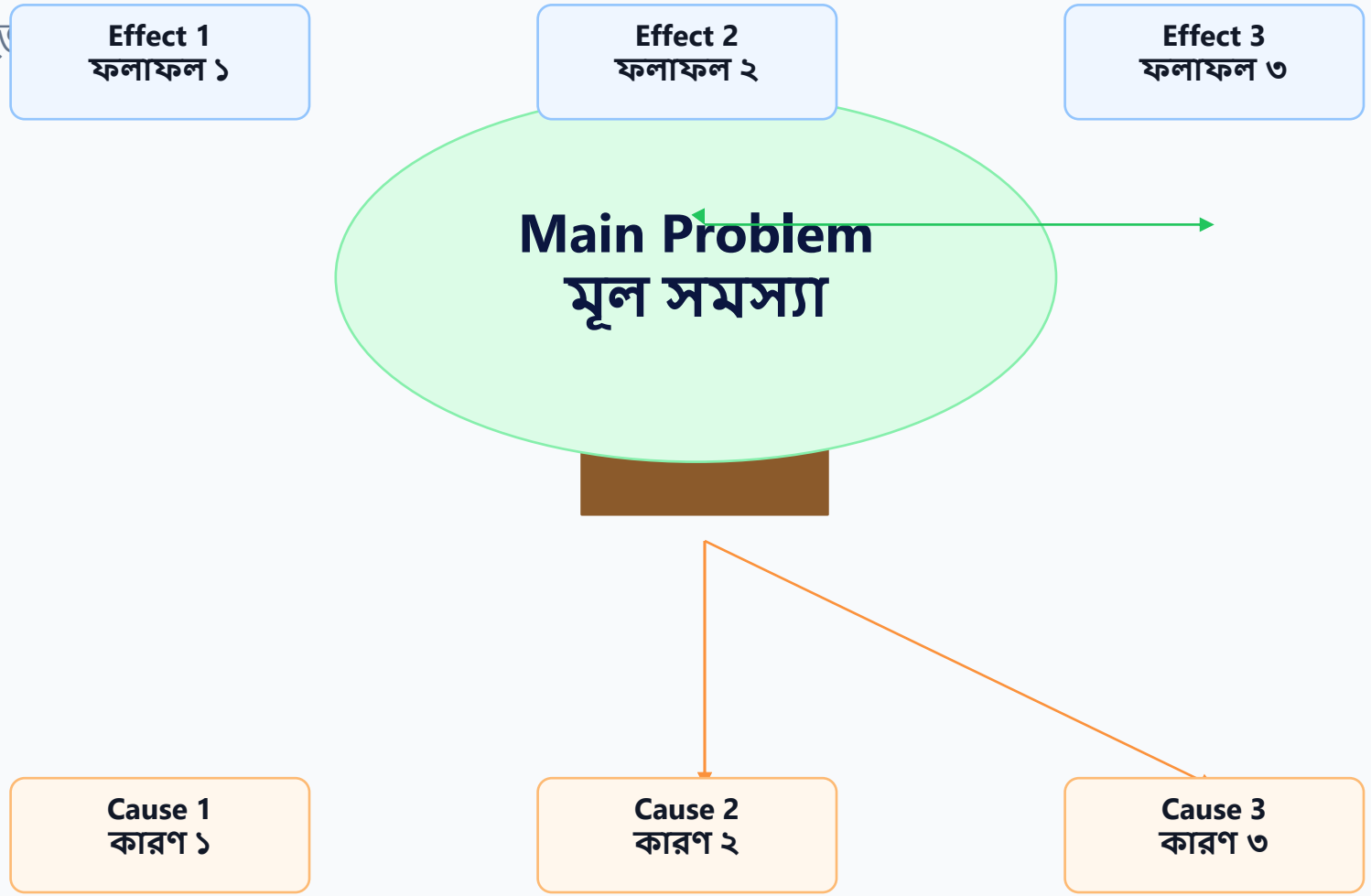
Bangla summary

৫W১H দিয়ে vague idea দ্রুত clear করা যায়।



Problem Tree Technique

সমস্যার root cause খুঁ



Use it when your topic feels too broad. Causes can become independent variables; effects can become outcomes.

Vague Topic to Clear Topic

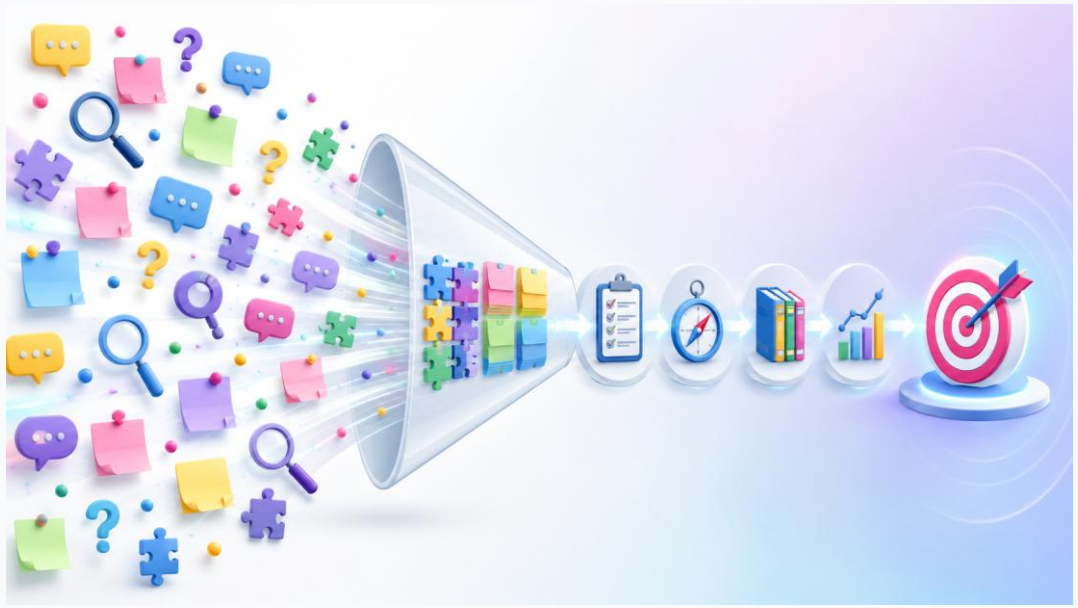
অস্পষ্ট topic থেকে পরিষ্কার research topic

Too broad	**Education**	শিক্ষা
Still broad	**AI in education**	শিক্ষায় AI
Better	**AI feedback and writing revision**	AI feedback ও writing revision
Researchable	**How first-year students use AI feedback to improve academic writing**	প্রথম বর্ষের শিক্ষার্থীরা academic writing উন্নত করতে AI feedback কীভাবে ব্যবহার করে

Always move from broad field to specific population, context, concept and evidence.

The 20 to 3 to 1 Idea Funnel

অনেক idea থেকে একটিকে বেছে নিন



20 raw ideas

যা মাথায় আসে লিখুন

→

3 shortlisted ideas

score দিয়ে shortlist করুন

→

1 final idea

সবচেয়ে feasible idea নিন

Selection rule

Final idea should be important, specific, ethical, feasible and supported by literature.

Idea Sources by Discipline

সব subject-এর জন্য example

English

Social media language and identity

Bangla

Translation quality in online news

CSE

Bias in Bangla fake news detection

MBBS

Patient waiting time and satisfaction

Engineering

Low-cost water filter performance

Business

Online review and purchase trust

Agriculture

Farmer adaptation to heat stress

Law

Digital privacy awareness among youth

Research logic is universal. Evidence type changes by discipline.

Check If Your Idea Is Already Done

আগে কেউ করেছে কি না দেখুন

Search keywords

মূল শব্দ লিখে Google Scholar বা database-এ খুঁজুন

Read recent papers

শেষ ৫ বছরের paper দেখুন

Compare context

দেশ, population, method, dataset আলাদা কি না দেখুন

Find uncertainty

যেখানে জানা নেই বা debate আছে সেটাই chance



Feasibility Test

আইডিয়া করা সম্ভব কি না যাচাই করুন

1. **Data**
   ডেটা পাওয়া যাবে?

2. **Access**
   participants, texts, labs, dataset পাওয়া যাবে?

3. **Time**
   সময় যথেষ্ট?

4. **Skills**
   analysis বা tools জানা আছে?

5. **Ethics**
   risk আছে কি? permission লাগবে?

6. **Supervisor fit**
   guide করার মতো mentor আছে?

A good idea is not only interesting. It must be doable.

Simple Idea Scoring Matrix

স্কোর দিয়ে idea বাছাই

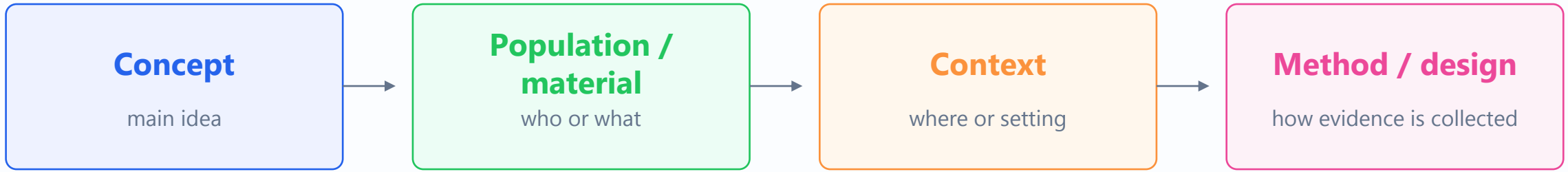
Idea	Importance	Novelty	Feasibility	Data	Total
AI feedback in writing	5	4	4	4	17
Crop disease app	4	4	2	2	12
Patient waiting time	5	3	5	5	18
Online review trust	4	3	5	5	17

How to use it

Give 1 to 5 score for each criterion. Select the idea with strong total score and clear personal interest.

Build a Strong Research Title

ভালো research title বানানোর formula



Example: AI feedback use and academic writing revision among first-year university students: a mixed-methods study

বাংলা: Concept, population, context এবং method থাকলে title দ্রুত পরিষ্কার হয়।

One Problem, Many Research Styles

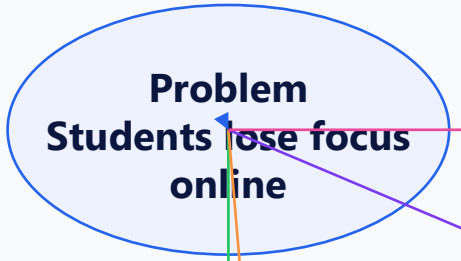
একটি problem থেকে অনেক ধরনের research

Survey

prevalence and factors

Text analysis

online discussion patterns



Interview

student experience

CSE system

focus tracking app

Experiment

test an intervention

Use AI as an Assistant, Not as the Researcher

AI সাহায্য করবে, চিন্তা আপনার

Good use

Brainstorm keywords, compare ideas, make outlines, simplify concepts.

Dangerous use

Inventing citations, copying full text, submitting AI claims without checking.

Best practice

Always verify with papers, data, supervisor and your own reasoning.



Rule: AI can suggest. Researcher must judge.

Common Beginner Mistakes

নতুনদের সাধারণ ভুল

Choosing too broad topic

অনেক বড় scope

Starting without literature

আগের কাজ না দেখে শুরু

No clear population or evidence

কার/কী নিয়ে কাজ তা অস্পষ্ট

Ignoring feasibility

সময়, data, permission না ভাবা

Copying title from paper

নিজস্ব angle নেই

Topic without method

কীভাবে answer করবেন জানা নেই

Fix: narrow the topic, check evidence, define context, choose a method and score feasibility.

30 Minute Class Activity

৩০ মিনিটের practice task

1

Minute 0 to 5

Write 10 everyday problems from your subject.

2

Minute 5 to 10

Select 3 problems that matter most.

3

Minute 10 to 15

Ask 5W1H questions for each.

4

Minute 15 to 20

Search quick keywords in Google Scholar.

5

Minute 20 to 25

Score novelty, importance and feasibility.

6

Minute 25 to 30

Write one final research title.

Final Recap

Everyday Problem to Research Idea

1. Observe real problems
2. Ask 5W1H questions
3. Check literature quickly
4. Test feasibility
5. Write a clear research title

Homework: Bring 3 scored ideas for Day 15.
হোমওয়ার্ক: Day 15-এর জন্য ৩টি scored idea তৈরি করুন।

Next: How to Choose a Strong and Feasible Research Topic

From Idea to Literature Review

আইডিয়া থেকে literature review table

1
Problem
সমস্যা

2
Question
প্রশ্ন

3
Evidence
প্রমাণ

4
Gap
গ্যাপ

5
Method
পদ্ধতি

Why this step matters

- Literature review shows what is already known.
- Knowledge gap explains why your study is needed.
- Method connects the gap with evidence.

Simple formula: Everyday Problem + Literature Evidence + Knowledge Gap + Feasible Method = Strong Research Topic

Example Topic and Search Plan

একটি topic ধরে literature search করার পদ্ধতি

Example topic

- AI feedback and academic writing revision among first-year university students
- Design idea: mixed-methods study

Search keywords

- AI feedback academic writing
- ChatGPT writing revision
- automated writing evaluation
- feedback engagement metacognition

Recent papers
শেষ ৫ বছরের paper

Compare context
population, country, method

Find gap
unknown / unclear / debated

Tip: do not copy a title. Use literature to build your own angle.

Literature Review Table: 5 Example Papers

৫টি paper সাজিয়ে evidence দেখা

Author	DOI	Method	Main finding	Gap
Mahapatra 2024	10.1186/s40561-024-00295-9	Mixed-method intervention	Improved academic writing; positive perceptions	Need genre & micro-writing evidence
Han & Li 2024	10.1016/j.system.2024.103502	AI + teacher feedback	More feedback incorporated across tasks	Need attitude & long-term outcome studies
Alsofyani & Barzanji 2025	10.1177/07356331241307297	Mixed-method comparison	Examined ChatGPT vs teacher feedback	Need multi-site ethical integration evidence
Teng 2024	10.1111/ejed.12811	Semester mixed-method	Metacognition shapes AI feedback use	Need engagement + revision behavior model
Lo et al. 2026	10.1016/j.caeai.2025.100511	AI feedback quality + interviews	Accurate but often generic; lacks examples	Need context-specific long-form feedback design

This table is an example. Replace with your own topic papers.

Evidence to Knowledge Gap

প্রমাণ থেকে research gap বানানো

Clear evidence

- AI feedback can improve writing performance.
- AI + teacher model gives broader feedback.
- Students value fast and available feedback.

Remaining gap

- Students may accept AI feedback blindly.
- Long-form writing feedback can be generic.
- Local context and revision process are underexplored.

Research opportunity

- Measure writing improvement.
- Track how feedback is used in revisions.
- Interview students about trust and decision-making.

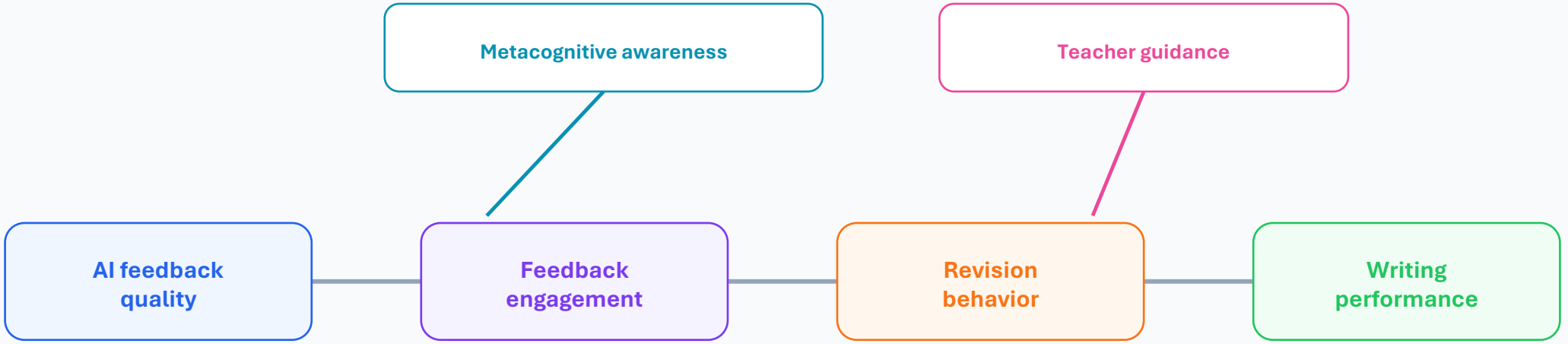
Example knowledge gap statement

Although AI feedback shows promise for academic writing, less is known about how students critically evaluate, trust, and incorporate AI feedback during revision in local university contexts.

বাংলা: gap মানে শুধু 'আগে হয়নি' নয়; gap মানে কী এখনও unclear বা underexplored।

Conceptual Framework

concept থেকে model বানানো



Model logic

- AI feedback must be understood before it improves writing.
- Engagement explains why feedback becomes revision.
- Teacher support reduces blind acceptance of AI suggestions.

Methods Model

research design and data collection

Design

Mixed-methods explanatory

Participants

First-year university students

Intervention

AI feedback on writing drafts

Quantitative data

Pre/post writing scores + survey

Qualitative data

Interviews about feedback use

Integration

Joint display of results

Main model: writing performance = f(AI feedback use, engagement, metacognition, baseline proficiency, background variables)

Variables and Measurement

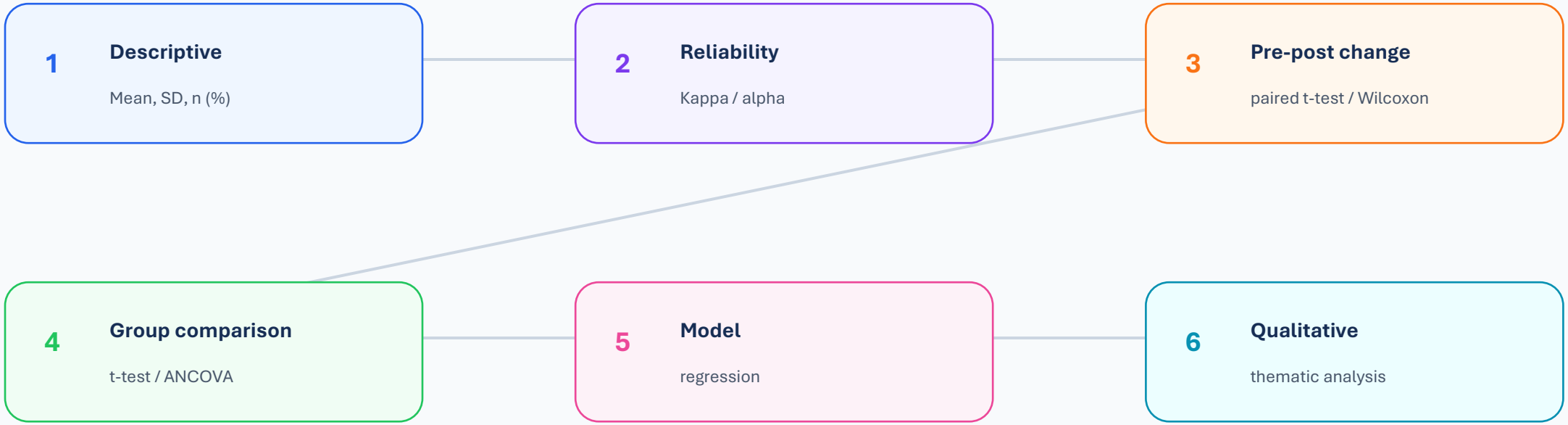
কোন variable কীভাবে মাপবেন

Outcome	**Writing score**	Rubric: content, organization, grammar, vocabulary
Outcome	**Revision quality**	accepted changes / quality of improvement
Exposure	**AI feedback use**	used or not used; frequency; prompt quality
Mediator	**Feedback engagement**	behavioral, cognitive, affective engagement
Moderator	**Metacognition**	planning, monitoring, evaluation
Controls	**Baseline proficiency**	pre-test score, age, gender, discipline

Tip: variables should come from the problem tree: causes = predictors, effects = outcomes.

Analysis Roadmap

descriptive থেকে final conclusion



Final output: evidence table + knowledge gap + conceptual framework + method + analysis plan

Class Activity: Make Your Own Review Table

নিজের topic দিয়ে practice করুন

10 min	Select 1 topic
15 min	Find 5 papers
15 min	Fill review table
10 min	Write knowledge gap
10 min	Draw model
10 min	Choose analysis plan

Homework: Bring one literature review table with 5 papers and one clear knowledge gap statement.